

Multichannel Visual Thermometer

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Temperature of some electronic components used in a device may rise unexpectedly and these might get damaged. This multichannel visual thermometer can indicate the problem through glowing LEDs. Since transistors are very cheap and widely available, this circuit uses these as low-cost temperature sensors and amplifiers.

Circuit and working

The multichannel visual thermometer (Fig. 1) is built around IC LM385 (IC1) and transistors BC550 (T1 through T4) and a few other compo-

nents. The circuit is based on the fact that voltage across the forward polarised emitter-base junction of the bipolar transistors is reduced with around $2.2\text{mV}/^{\circ}\text{C}$. IC1 produces a reference voltage of around 1.2V.

The transistors can work as temperature sensors and amplifiers-cum-comparators at the same time. These should be mounted with very good thermal contact over the monitored components (transistors, heat-sinks, diodes, transformers, power cables, heating elements, etc).

Adjustment of the circuit is simple. At room temperature (say, around 25°C), adjust potmeters in such a way that there is no visible

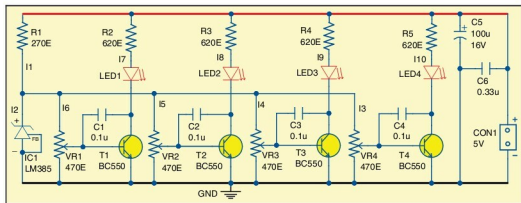


Fig. 1: Circuit diagram of the visual thermometer

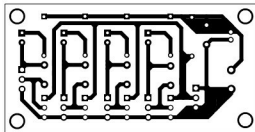


Fig. 2: Actual-size PCB of the thermometer

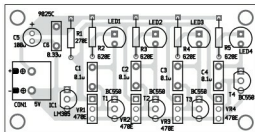


Fig. 3: Component layout of the PCB

light from the LEDs.

When you heat sensors T1 through T4, LEDs (LED1 through LED4) should start emitting light in proportion to the temperature of the corresponding transistors.

For testing, you can heat (usually below 150°C) the transistors for a short time with a soldering iron. You can adjust different thresholds depending on the temperature of interest, with the LEDs emitting light. The 5V DC power supply is connected across CON1.

LM385 has a maximal current of 20mA, which should be taken into consideration when

values of R1 and VR1 through VR4 are calculated. Current I2 (see Fig. 1) should always be between $10\mu\text{A}$ and

20mA. Currents I3 to I6 should be at least ten times higher than the required maximal base current for transistors T1 through T4. Currents I7 through I10 should be minimal for producing light in LEDs at temperatures of interest, for example, between 5mA and 10mA. Capacitors C1 through C4 prevent oscillation from transistors and may be omitted.

Construction and testing

An actual-size, single-side PCB for the visual thermometer is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a suitable small box in such a way that transistors T1 through T4 absorb the heat when installed in a suitable instrument. Conductors between the transistors and the PCB should be as short as possible, and no longer than 50cm. ●

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PARTS LIST

Semiconductors:	
IC1	- LM385 voltage reference diode
T1-T4	- BC550 npn transistor
LED1-LED4	- 5mm LED
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1	- 270-ohm
R2-R5	- 620-ohm
VR1-VR4	- 470-ohm potmeter
Capacitors:	
C1-C4	- 0.1 μF ceramic disk
C5	- 100 μF , 16V electrolytic
C6	- 0.33 μF ceramic disk
Miscellaneous:	
CON1	- 2-pin connector terminal
	- 5V DC power supply